

Numerical Diagnostics for Topological Frustration in a Domain-Wall-Truncated Adaptive Fermion Circuit

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We specify a CPU reference program that distinguishes three logically separate features of a number-conserving adaptive Gaussian circuit: algebraic compatibility of its target measurements, stochastic defect and transfer activity along trajectories, and slow single-particle dynamics near a hard interface. The interface is produced by truncating every controller mode to the sector containing its center and by updating only the active topological slab. It therefore does not require a physically trivial exterior mass. The reference implementation nevertheless retains the canonical geometry parameters $\alpha_1 = 1$ and $\alpha_2 = 30$. We define exact Gaussian-completion tests, a weighted minimum-frustration problem, defect and signed-transfer records, empirical full-counting statistics, two deliberately distinct spectral proxies, and a paired local-response test for directional wall propagation. The note also states the numerical controls and the inference licensed by each observable. It is a methods and validation document and contains no numerical claims.

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I. CONTROLLER AND HARD-WALL GEOMETRY

The active single-particle Hilbert space has dimension $D = 2N_{\text{cell}}$, with two complex orbitals per cell. A covariance $G = 2C - \mathbf{1}$ is represented in terms of the one-body correlation matrix $0 \leq C \leq \mathbf{1}$. A pure number-conserving Gaussian state at half filling corresponds to a rank- N projector C , where $N = D/2$.

At each cell $R = (R_x, R_y)$ the controller contains four normalized observable Wannier vectors

$$|w_{aR}\rangle, \quad a \in \{A+, A-, B+, B-\}, \quad (1)$$

with desired occupations

$$s_{A+} = s_{B+} = 0, \quad s_{A-} = s_{B-} = 1. \quad (2)$$

The untruncated vectors are obtained from the upper- and lower-band projectors of the flattened class-A Hamiltonian at $\alpha = 1$. A finite support cutoff n_{shell} is applied first. For a domain-wall center R , the additional binary mask is

$$M_R(r) = \begin{cases} 1, & r \text{ and } R \text{ lie in the same domain-wall sector,} \\ 0, & \text{otherwise,} \end{cases} \quad |w_{aR}^{\text{DW}}\rangle = \frac{M_R|w_{aR}\rangle}{\|M_R|w_{aR}\rangle\|}. \quad (3)$$

Only centers and one-body indices inside the slab $x_L \leq x \leq x_R$ are retained by `meas_slab_only`. Consequently, active controller vectors are determined entirely by α_1 and the mask in Eq. (3). Changing the unused exterior value α_2 changes only excluded exterior frames. The production configuration is therefore

$$\alpha_1 = 1, \quad \alpha_2 = 30, \quad \text{DW}=\text{True}, \quad \text{dw_truncation}=\text{True}, \quad \text{meas_slab_only}=\text{True}. \quad (4)$$

An automated invariance test compares active frames at $\alpha_2 = 1$ and 30; this comparison is a software control, not a physics sweep.

II. STATIC GAUSSIAN COMPLETION

Let W_F contain the desired-filled vectors $A-, B-$ and W_E the desired-empty vectors $A+, B+$. Rank-revealing singular-value decompositions, with relative tolerance 10^{-10} , define orthonormal bases Q_F, Q_E and subspaces $\mathcal{V}_F, \mathcal{V}_E \subset \mathcal{H}$. The singular values of $Q_F^\dagger Q_E$ are $\cos \vartheta_j$. We report

$$\sigma_{\max} = \max_j \cos \vartheta_j, \quad \Phi = \text{Tr}(P_F P_E) = \sum_j \cos^2 \vartheta_j, \quad (5)$$

where $P_F = Q_F Q_F^\dagger$ and $P_E = Q_E Q_E^\dagger$.

Proposition 1 (Exact completion). *A rank- N projector C satisfying $C|w\rangle = |w\rangle$ for every filled constraint and $C|w\rangle = 0$ for every empty constraint exists if and only if*

$$\mathcal{V}_F \perp \mathcal{V}_E, \quad \text{rank } W_F \leq N \leq D - \text{rank } W_E. \quad (6)$$

Indeed, necessity follows by placing $\mathcal{V}_F$ in $\text{im } C$ and $\mathcal{V}_E$ in $\text{ker } C$. Conversely, under Eq. (6), augment $\mathcal{V}_F$ by any $N - \dim \mathcal{V}_F$ directions in $(\mathcal{V}_F \oplus \mathcal{V}_E)^\perp$ and project onto the resulting space. The code constructs this projector when possible and checks filled leakage, empty leakage, and rank error.

When exact completion fails, define the weighted defect functional

$$\mathcal{F}[C] = \sum_a \gamma_a [s_a + (1 - 2s_a) \langle w_a | C | w_a \rangle], \quad \text{rank } C = N, \quad (7)$$

with $\gamma_a = 1$ by default. Writing

$$A = \sum_a \gamma_a (1 - 2s_a) |w_a\rangle \langle w_a|, \quad (8)$$

Ky Fan's variational principle gives

$$F_\star = \min_C \mathcal{F}[C] = \sum_a \gamma_a s_a + \sum_{j=1}^N \lambda_j^\dagger(A). \quad (9)$$

The minimizing eigenspace yields channel-resolved residuals $f_{a\star}$ and the map $f_\star(x, y)$. Equations (5)–(9) answer whether the controller constraints admit a common Gaussian state and where an irreducible static compromise resides. They do not determine whether trajectory activity survives at long times.

III. TRAJECTORY ACTIVITY AND COUNTING STATISTICS

For the m th visited channel, let $n_{a,m} \in \{0, 1\}$ be the sampled Born-rule outcome and $n'_{a,m}$ the occupation imposed by any subsequent feedback reset. We record

$$X_{a,m} = |n_{a,m} - s_a|, \quad Y_{a,m} = n'_{a,m} - n_{a,m} \in \{-1, 0, +1\}. \quad (10)$$

Thus X is a wrong-outcome indicator, $|Y|$ is actual transfer activity, and the sign of Y distinguishes gain from loss. Under perfect correction, $X = |Y|$ event by event. Under imperfect feedback they need not agree. Forced postselection does not produce a stochastic activity record and is never encoded as a string of zero events. The desired-outcome Born probability is retained as a conditional frustration diagnostic.

The interface region contains the active columns within $w_{\text{int}} = n_{\text{shell}} + 1$ of either hard wall; the remaining active cells form the interior. For a region R and observation window T , define

$$K_R^X(T) = \sum_{m \in R, T} X_m, \quad K_R^Y(T) = \sum_{m \in R, T} |Y_m|. \quad (11)$$

The code reports rates, spatial profiles, local inter-event survival probabilities, and the first three cumulants divided by T . Its finite-time, empirical scaled cumulant-generating function is

$$\theta_R(s, T) = \frac{1}{T} \log \left\langle e^{-s K_R(T)} \right\rangle_{\text{traj}}. \quad (12)$$

Evaluation uses log-sum-exp stabilization on 41 fields in $-0.2 \leq s \leq 0.2$, with bootstrap intervals and a reweighting effective sample fraction. Points below 0.1 effective sample fraction are flagged. Equation (12) measures finite-time fluctuations and intermittency; it is not the leading eigenvalue of an explicitly constructed tilted channel, and finite-size curvature is not a proof of a dynamical singularity.

IV. SLOW-MODE PROXIES

A. Lyapunov tangent cocycle

Along a sampled trajectory, the existing engine propagates a one-body tangent frame V_t through the differential of every measurement and reset. A QR factorization after each cycle gives

$$V_t = Q_t R_t, \quad \lambda_j(t) = \frac{1}{t} \sum_{\tau=1}^t \log |(R_\tau)_{jj}|. \quad (13)$$

We retain the full active-space spectrum on small systems and define

$$\Delta_L(t) = \min_j |\lambda_j(t)|. \quad (14)$$

The final minimum- $|\lambda|$ vector is characterized by its interface weight, left/right-wall weights, inverse participation ratio, and transverse profile. This diagnoses marginal directions of the implemented single-particle tangent cocycle, not the full nonlinear stochastic channel.

B. Regularized-Choi particle transfer

A separate run propagates the regularized Gaussian Choi covariance. On its finite sector, the particle transfer matrix is

$$T_p = -(\mathbf{1} + \Sigma_{LL})^{-1} \Sigma_{LR}. \quad (15)$$

For a pure Choi covariance, diagonalizing the Hermitian block Σ_{LL} with eigenvalues $a_j \in [-1, 1]$ gives finite transfer exponents

$$\xi_j(t) = \frac{1}{2t} [\log(1 - a_j) - \log(1 + a_j)], \quad \Delta_{\text{Choi}}(t) = \min_{j \text{ finite}} |\xi_j(t)|. \quad (16)$$

Endpoint zeros and poles, spectral-interval violations, eigensystem residuals, and near-gap eigenvectors are saved explicitly. If no finite sector remains, Δ_{Choi} is undefined rather than zero; the latest finite near-gap vectors and their observation cycle are retained. The Lyapunov and Choi gaps are compared side by side but never merged: they answer different linearized questions and neither is labeled the exact Markov-channel gap.

V. PAIRED DIRECTIONAL RESPONSE

After $2N_y$ equilibration cycles, select a unit cell (x_w, y_0) on a wall and let P project onto its two onsite orbitals. From the same equilibrated covariance construct

$$G^+ = QGQ + P, \quad G^- = QGQ - P, \quad Q = \mathbf{1} - P. \quad (17)$$

The pair is evolved with identical per-sample random streams and identical randomized schedules under perfect correction. Because this protocol consumes one Born variate per channel and no stochastic correction variate, the common stream remains aligned even when paired outcomes differ. The normalized response is

$$\delta n(x, y, t) = \frac{n^+(x, y, t) - n^-(x, y, t)}{2}, \quad \sum_{x, y} \delta n(x, y, 0) = 1. \quad (18)$$

After projecting onto each wall strip, the code records the response norm, absolute width, wall weight, and periodic signed moment

$$M_1(t) = \frac{\sum_y d_y \delta n_{\text{wall}}(y, t)}{\sum_y |\delta n_{\text{wall}}(y, t)|}, \quad d_y \in [-N_y/2, N_y/2]. \quad (19)$$

An early-time slope estimates directed drift before periodic wraparound. Random update order is primary; raster order quantifies ordering bias. A credible chiral response requires wall localization, opposite drift signs at the two oppositely oriented walls, stability under randomized order, and no analogous stable signal in the uniform controller.

VI. VALIDATION AND NUMERICAL PROGRAM

TABLE I. Question, control, and licensed inference for each diagnostic.

Observable	Required test or control	Information supplied
$\sigma_{\max}, \Phi$	Synthetic orthogonal and overlapping spans; rank threshold stability	Exact mutual compatibility of filled and empty target subspaces
$F_\star, f_\star(x, y)$	Ky Fan formula versus direct residual sum; α_2 invariance	Minimum static defect cost and its location, but not persistent activity
X, Y	Live branch parser; $X = Y $ under perfect correction; imperfect-feedback control	Wrong outcomes versus actual particle transfer
$\theta_R(s, T)$	Bernoulli analytic benchmark, $\theta(0, T) = 0$, bootstrap and effective sample size	Finite-time activity fluctuations and intermittency, not an exact tilted-channel phase transition
Δ_L	Dense small-system spectrum and final-vector localization	Slow or marginal one-body tangent directions
Δ_{Choi}	Known Σ_{LL} spectra, endpoint counts, eigenpair residuals	Slow regularized one-body particle-transfer sectors, distinct from Δ_L
$M_1(t)$	Common seeds, random-order primary run, raster sensitivity, uniform control	Direction and localization of a wall response; opposite wall velocities provide the chirality test

The smoke suite uses 4×6 uniform and hard-wall systems. The reference activity grid fixes $N_x = 12$, takes $N_y = 16, 24, 32$, uses 64 trajectories and $4N_y$ cycles, and discards the first $2N_y$. Spectral reference runs use $N_x = 8$, $N_y = 8, 12, 16$, eight trajectories, and $2N_y$ cycles. Response runs use the activity geometries, 64 paired trajectories, $2N_y$ equilibration cycles, and an $N_y/2$ response horizon. The default cutoff is $n_{\text{shell}} = 1$. Runs at $n_{\text{shell}} = 2$ are explicit dynamical convergence controls, and the untruncated frame is an explicit static-only control; neither is launched by default.

Every output stores the canonical entry point, geometry, active basis, tolerances, root seed, derived sample seeds, protocol, and update order. Equilibration observers stream charge, charge variance, Gaussian entropy, purity defect, and successive covariance change. No full covariance history is required. Independent geometries and trajectories execute in CPU process workers, with numerical-library threads capped per worker. Worker seeds are derived before dispatch, results are merged in task order, and only the parent process writes files. The run record includes the CPU and memory budget, chosen worker and thread counts, process identifiers, and task timing. Production characterization begins only after unit tests, α_2 invariance, serial-parallel reproducibility, the full small-system CLI smoke campaign, and a measured parallel-utilization benchmark pass. The launcher then runs static, activity, and response pilots before their production stages; any failed gate terminates the sequence without a success marker.

VII. INTERPRETIVE LIMITS

The hard mask deliberately studies a controller boundary rather than a smooth material interface. Smooth walls and mixed topological/trivial projector families are outside the present scope. A nonzero $F_\star$ establishes an algebraic compromise but does not force a nonzero stationary rate. A finite empirical SCGF samples trajectory fluctuations but does not replace a tilted operator. Closing Lyapunov or Choi proxies are evidence for slow one-body sectors only after localization and finite-size checks. Directional response becomes evidence for chirality only after update-order and uniform controls. These separations are part of the numerical specification, not merely terminological cautions.